

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

CAMERA CONTROL MODULE (CCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Camera Control Module (CCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Parking Aid](#) (413-13 Parking Aid,

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B100E-29	Video Input A signal - Signal invalid	Video out circuit short circuit to ground, open circuit, high resistance	Refer to the electrical circuit diagrams and check the video out circuit for short circuit to ground, open circuit, high resistance
B108E-31	Display - No signal	- Missing/invalid data from the touch screen - Camera control module internal failure	- Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new camera control module
B12BD-18	Rear Camera - Circuit current below threshold	- Rear proximity camera power or ground circuit open circuit, high resistance - Rear proximity camera internal failure	- Refer to the electrical circuit diagrams and check the rear proximity camera power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear proximity camera
B12BD-19	Rear Camera - Circuit current above threshold	- Rear proximity camera circuit short circuit to ground, short circuit to power - Rear proximity camera internal failure	- Refer to the electrical circuit diagrams and check the rear proximity camera circuit for short circuit to ground, short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear proximity camera
B12BD-31	Rear Camera - No signal	Rear proximity camera low	Refer to the electrical circuit diagrams and check the rear

		voltage differential signalling circuits short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear proximity camera internal failure	proximity camera low voltage differential signalling circuits for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear proximity camera
B12BD-49	Rear Camera - Internal electronic failure	▪ Rear proximity camera internal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new rear proximity camera
B12BE-18	Left Front Camera - Circuit current below threshold	▪ Front proximity camera power or ground circuit open circuit, high resistance ▪ Front proximity camera internal failure	▪ Refer to the electrical circuit diagrams and check the front proximity camera power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front proximity camera
B12BE-19	Left Front Camera - Circuit current above threshold	▪ Front proximity camera circuit short circuit to ground, short circuit to power ▪ Front proximity camera internal failure	▪ Refer to the electrical circuit diagrams and check the front proximity camera circuit for short circuit to ground, short circuit to power ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front proximity camera
B12BE-31	Left Front Camera - No signal	▪ Front proximity camera low voltage differential signalling circuits short circuit to	▪ Refer to the electrical circuit diagrams and check the front proximity camera low voltage differential signalling circuits for short circuit to ground, short circuit to power, open circuit, high resistance

		ground, short circuit to power, open circuit, high resistance Front proximity camera internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front proximity camera
B12BE-49	Left Front Camera - Internal electronic failure	Front proximity camera internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new front proximity camera
B12C0-18	Left Mirror Camera - Circuit current below threshold	- Left mirror proximity camera power or ground circuit open circuit, high resistance - Left mirror proximity camera internal failure	- Refer to the electrical circuit diagrams and check the left mirror proximity camera power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new left mirror proximity camera
B12C0-19	Left Mirror Camera - Circuit current above threshold	- Left mirror proximity camera circuit short circuit to ground, short circuit to power - Left mirror proximity camera internal failure	- Refer to the electrical circuit diagrams and check the left mirror proximity camera circuit for short circuit to ground, short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new left mirror proximity camera
B12C0-31	Left Mirror Camera - No signal	Left mirror proximity camera low voltage differential signalling circuits short circuit to ground, short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check the left mirror proximity camera low voltage differential signalling circuits for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

		Left mirror proximity camera internal failure	left mirror proximity camera
B12C0-49	Left Mirror Camera - Internal electronic failure	Left mirror proximity camera internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new left mirror proximity camera
B12C1-18	Right Mirror Camera - Circuit current below threshold	- Right mirror proximity camera power or ground circuit open circuit, high resistance - Right mirror proximity camera internal failure	- Refer to the electrical circuit diagrams and check the right mirror proximity camera power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new right mirror proximity camera
B12C1-19	Right Mirror Camera - Circuit current above threshold	- Right mirror proximity camera circuit short circuit to ground, short circuit to power - Right mirror proximity camera internal failure	- Refer to the electrical circuit diagrams and check the right mirror proximity camera circuit for short circuit to ground, short circuit to power - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new right mirror proximity camera
B12C1-31	Right Mirror Camera - No signal	- Right mirror proximity camera low voltage differential signalling circuits short circuit to ground, short circuit to power, open circuit, high resistance - Right mirror proximity camera internal	- Refer to the electrical circuit diagrams and check the right mirror proximity camera low voltage differential signalling circuits for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new right mirror proximity camera

		failure	
B12C1-49	Right Mirror Camera - Internal electronic failure	Right mirror proximity camera internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new right mirror proximity camera
U0011-81	Medium Speed CAN Communication Bus - Invalid serial data received	Invalid data received from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-87	Medium Speed CAN Communication Bus - Missing message	Missing message from another control module via the medium speed CAN bus (comfort)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0011-88	Medium Speed CAN Communication Bus - Bus off	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Camera control module is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the camera control module with the latest level software
U0400-29	Invalid Data Received - Signal invalid	Missing/invalid data from the touch screen	Using the manufacturer approved diagnostic system, check the touch screen for related DTCs and refer to the

		▪ Missing/invalid data from the anti-lock brake system control module ▪ Missing/invalid data from the steering angle sensor module	relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the steering angle sensor module for related DTCs and refer to the relevant DTC index
U0439-29	Invalid Data Received From All Terrain Control Module - Signal invalid	▪ Missing/invalid data from the transmission control switch	▪ Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index
U0447-29	Invalid Data Received From Gateway A - Signal invalid	▪ Missing/invalid data from the gateway module ▪ Camera control module power circuit short circuit to power	▪ Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the camera control module power circuit for short circuit to power
U044A-29	Invalid Data Received From Gateway D - Signal invalid	▪ Missing/invalid data from the transmission control switch ▪ Missing/invalid data from the anti-lock brake system control module	▪ Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0533-29	Invalid Data Received From Side Obstacle	▪ Missing/invalid data from the	▪ Using the manufacturer approved diagnostic system,

	Detection Control Module-Left - Signal invalid	left blindspot monitoring system control module	check the left blindspot monitoring system control module for related DTCs and refer to the relevant DTC index
U0534-29	Invalid Data Received From Side Obstacle Detection Control Module-Right - Signal invalid	Missing/invalid data from the right blindspot monitoring system control module	Using the manufacturer approved diagnostic system, check the right blindspot monitoring system control module for related DTCs and refer to the relevant DTC index
U1A4B-49	Control Module Processor B - Internal electronic failure	Missing/invalid data within the camera control module	Using the manufacturer approved diagnostic system, re-configure the camera control module with the latest level software
U201A-51	Control Module Main Calibration Data - Not programmed	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2023-41	Control Module Network Signal Calibration Data - General checksum failure	Camera control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new camera control module
U2100-00	Initial Configuration Not Complete - No sub type information	Car configuration file mismatch with vehicle specification	

			NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2101-00	Control Module Configuration Incompatible - No sub type information	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary
U2300-54	Central Configuration - Missing calibration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-55	Central Configuration - Not configured	Car configuration	

		file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-56	Central Configuration - Invalid/incomplete configuration	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest
U2300-64	Central Configuration - Signal plausibility failure	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the DTCs Using the manufacturer approved diagnostic system, check and update the car configuration file as necessary. Clear the DTCs and retest

U3000-46	Control Module - Calibration/parameter memory failure	Camera control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new camera control module
U3000-49	Control Module - Internal electronic failure	Camera control module internal failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new camera control module
U3000-98	Control Module - Component or system over temperature	Component or system over temperature	NOTE: Consider environmental conditions before condemning the camera control module Allow the camera control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new camera control module
U3006-16	Control Module Input Power A - Circuit voltage below threshold	- Camera control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Refer to the electrical circuit diagrams and check the camera control module power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system
U3006-17	Control Module Input Power A - Circuit voltage above threshold	Battery/charging system fault	Refer to the relevant section of the workshop manual and test the battery and charging system

